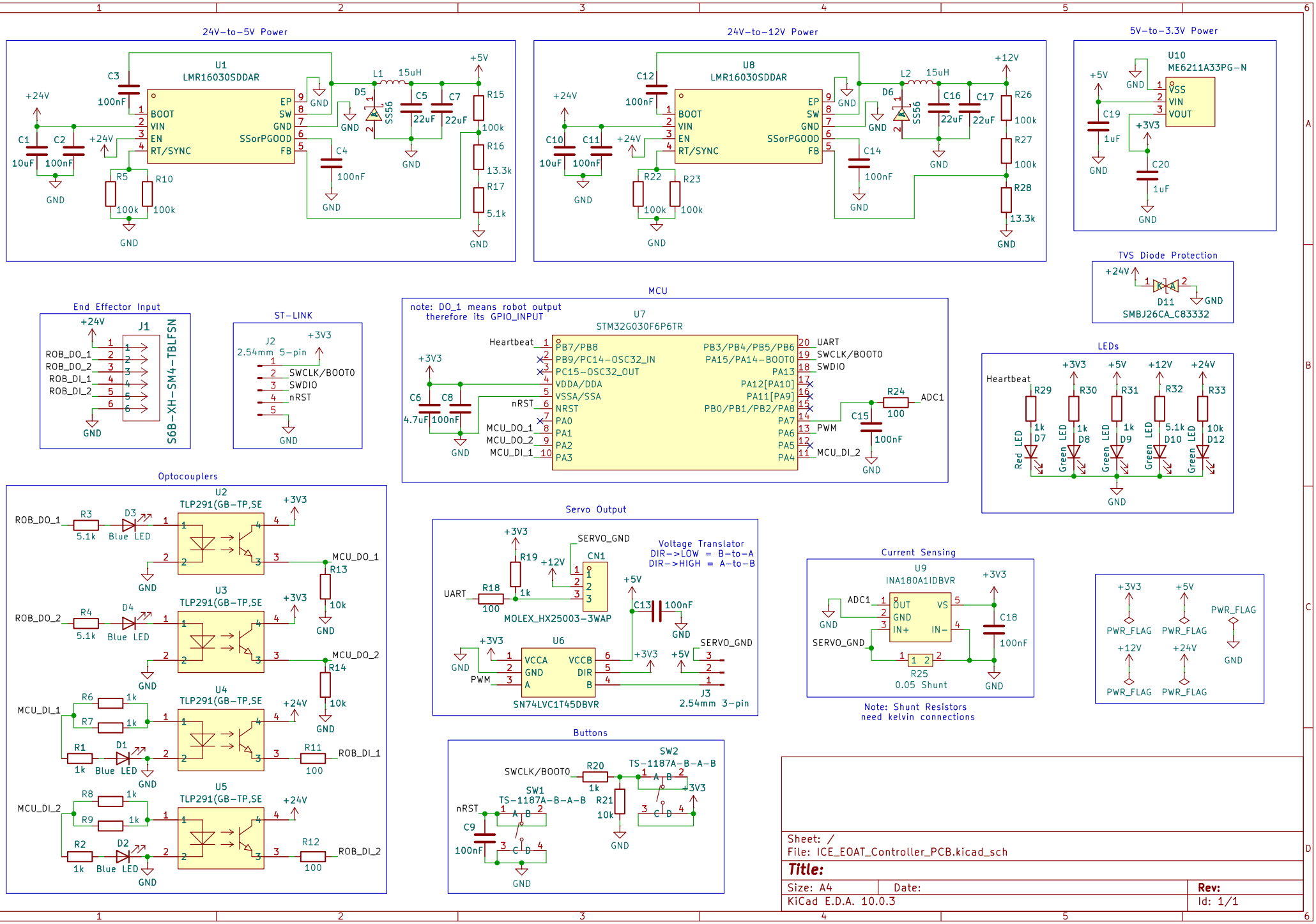

24V-to-5V Power
U1
LMR16030SDDAR
BOOT
VIN
EN
RT/SYNC
EP
SW
GND
SSorPGOOD
FB
C3
100nF
C1
10uF
C2
100nF
R5
100k
R10
100k
L1
15uH
D5
SS56
C5
22uF
C7
22uF
C4
100nF
R15
100k
R16
13.3k
R17
5.1k
+24V
+5V
GND
24V-to-12V Power
U8
LMR16030SDDAR
C12
100nF
C10
10uF
C11
100nF
R22
100k
R23
100k
L2
15uH
D6
SS56
C16
22uF
C17
22uF
C14
100nF
R26
100k
R27
100k
R28
13.3k
+12V
5V-to-3.3V Power
U10
ME6211A33PG-N
VSS
VIN
VOUT
C19
1uF
C20
1uF
+3V3
TVS Diode Protection
D11
SMBJ26CA_C83332
End Effector Input
J1
S6B-XH-SM4-TBLFSN
ROB_DO_1
ROB_DO_2
ROB_DI_1
ROB_DI_2
ST-LINK
J2
2.54mm 5-pin
SWCLK/BOOT0
SWDIO
nRST
MCU
note: DO_1 means robot output
therefore its GPIO_INPUT
U7
STM32G030F6P6TR
Heartbeat
PB7/PB8
PB9/PC14-OSC32_IN
PC15-OSC32_OUT
VDDA/DDA
VSSA/SSA
NRST
PA0
PA1
PA2
PA3
PB3/PB4/PB5/PB6
PA15/PA14-BOOT0
PA13
PA12[PA10]
PA11[PA9]
PB0/PB1/PB2/PA8
PA7
PA6
PA5
PA4
UART
SWCLK/BOOT0
SWDIO
PWM
MCU_DO_1
MCU_DO_2
MCU_DI_1
MCU_DI_2
ADC1
C6
4.7uF
C8
100nF
R24
100
C15
100nF
LEDs
Heartbeat
R29
1k
D7
Red LED
R30
1k
D8
Green LED
R31
1k
D9
Green LED
R32
5.1k
D10
Green LED
R33
10k
D12
Green LED
Optocouplers
U2
TLP291(GB-TP,SE
U3
TLP291(GB-TP,SE
U4
TLP291(GB-TP,SE
U5
TLP291(GB-TP,SE
ROB_DO_1
ROB_DO_2
MCU_DI_1
MCU_DI_2
R3
5.1k
R4
5.1k
R6
1k
R7
1k
R8
1k
R9
1k
R1
1k
R2
1k
D3
Blue LED
D4
Blue LED
D1
Blue LED
D2
Blue LED
R13
10k
R14
10k
R11
100
R12
100
ROB_DI_1
ROB_DI_2
Servo Output
SERVO_GND
Voltage Translator
DIR->LOW = B-to-A
DIR->HIGH = A-to-B
R19
R18
100
1k
UART
CN1
MOLEX_HX25003-3WAP
C13
100nF
U6
VCCA
GND
A
VCCB
DIR
B
SN74LVC1T45DBVR
PWM
J3
2.54mm 3-pin
Current Sensing
U9
INA180A1IDBVR
ADC1
OUT
GND
IN+
VS
IN-
SERVO_GND
C18
100nF
R25
0.05 Shunt
Note: Shunt Resistors
need kelvin connections
PWR_FLAG
Buttons
SW1
TS-1187A-B-A-B
SW2
TS-1187A-B-A-B
SWCLK/BOOT0
nRST
R20
1k
R21
10k
C9
100nF
Sheet: /
File: ICE_EOAT_Controller_PCB.kicad_sch
Title:
Size: A4
Date:
Rev:
KiCad E.D.A. 10.0.3
Id: 1/1